Next-Gen Smart Healthcare Using UAV Assisted Cooperative Communication: A Deep Learning Approach

Sunil Pattepu, Amlan Datta, Bhargav Appasani, Mohammad S. Khan, *Senior Member, IEEE*, Amrit Mukherjee, *Senior Member, IEEE*, and Milan Novak

***Abstract*—Unmanned aerial vehicles (UAVs) are evolving rapidly and are revolutionizing the operation of several applications. An important application that can be significantly improved with UAV-assisted communication is healthcare. Wearable devices can be worn on the body and are equipped with sensors that can detect and monitor various physiological parameters. The data collected by these devices can be used to track changes in health status over time, identify potential health risks, and inform clinical decision-making. UAV-assisted non-terrestrial wireless communication networks can play a significant role in the reliable transfer of this data. The novelty of the work is that it considers multi-antenna UAVs for communicating patient data collected from various wearable devices. It also proposes a hybrid relaying scheme incorporating deep learning (DL) for improved performance. The UAVs have pre-trained deep neural networks (DNNs) to process patient data and make accurate decisions without forwarding it to the healthcare facility. A two-layer deep neural network has been used at the UAV with a training accuracy of 99.6%, validation accuracy of 96.6%, and test accuracy of 96.62%. The UAV relays the data to the healthcare facility for expert opinion if it cannot make decisions with the desired accuracy. In this scenario, the optimum UAV and transmit antenna are selected, and a hybrid relaying scheme is used to minimize the network outage and maximize the throughput. The communication model is built, and extensive simulations are performed using patient data to demonstrate the approach. The work will significantly impact the use of UAVs for smart healthcare.**

***Index Terms*—Consumer electronic devices, cooperative communication, non-terrestrial-network, smart healthcare systems, deep learning, multi-antenna UAVs network.**

I. Introduction

Unmanned Aerial Vehicle (UAV) technology is evolving in recent years with developments in communication and computational paradigms. UAVs are pervading various applications and are revolutionizing their operation. Healthcare is one such sector that is witnessing the potential of UAVs in making it accessible to patients. Combined with cooperative communication and deep learning (DL), the UAVs can act as decision support systems (DSS) for providing real-time diagnostics, health monitoring, and disease prediction to remotely connected users, as illustrated in Fig 1. In Fig 1, it can be observed that the patients are remotely located and have no direct access to expert healthcare. They have several wearable devices such as smart watches, smart sensors, etc., that capture vital bodily parameters. This data captured by the wearable devices can be sent to the doctors for immediate diagnosis and medical help. Terrestrial networks are not always available for establishing communication links between the patient (equipped with wearable electronic devices) and the medical expert. Cooperative communication using UAVs with multiple antennas can establish non-terrestrial networks (NTNs) for reliably relaying the data generated by wearable devices over long distances. NTN is a communication technology that uses satellites and other non-terrestrial vehicles to bring connectivity to regions previously unreachable by terrestrial networks, whether over mountains, across deserts, or in the middle of the ocean. NTNs consist of variants of space-borne and aerial communication networks, including GEO, MEO, LEO satellite constellations, High Altitude Platform Systems (HAPS), Low Altitude Platform Systems (LAPS), and air-to-ground (A2G) networks. Multi-antenna devices UAVs have emerged as capable NTN expansion devices and have garnered significant interest due to numerous applications

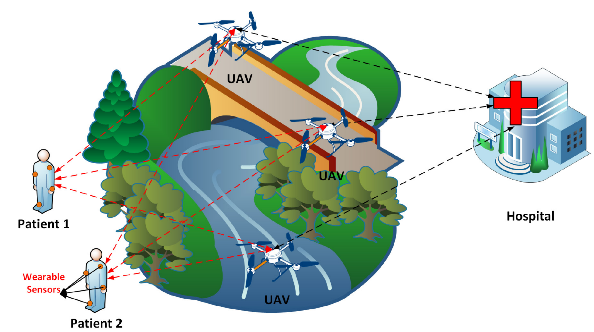

Fig. 1. UAVs Assisted Next-G Smart Healthcare System with Data Collected from Wearable Devices.

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in fifth-generation (5G) cooperative networks [1]. UAVs are being considered in real-world scenarios for relaying patient data to hospitals, such as with the MOMENTUM project [2]. As UAVs are being used by patients to relay their data, patients can also be considered consumers, and UAVs can be considered consumer electronic devices [3], [4]. Additionally, implementing machine learning (ML) and DL algorithms in multi-antenna systems can increase the quality of service in health monitoring systems. The motivation of this research is to utilize these technologies for smart healthcare applications.

A. Related Work

In cooperative communication networks, amplify and forward (AF) and decode and forward (DF) are widely used for relaying [5]. AF relaying involves the relay amplifying the received signal before forwarding it, while DF relaying requires the relay to decode the received signal before retransmitting it. The choice between AF and DF affects the relay transmission's outage probability and symbol error rate (SER). Hybrid relaying schemes have been shown to significantly improve system performance compared to traditional relaying schemes in terms of outage probability, error performance, and link efficiency [6], [7], [8]. These schemes involve relay nodes making decisions based on signal-to-noise ratio (SNR) thresholds to operate in DF, AF, or silent modes [9]. A hybrid relay selection algorithm, NFHDAF, improves the system's performance by switching between DF or AF mode to achieve a lower outage probability [10]. A hybrid decode-amplify-forward (HDAF) method for half-duplex relaying networks has been proposed to reduce outage probability and achieve a higher average capacity than individual AF and DF schemes [8]. However, these earlier works on cooperative communication do not consider UAVs as relays.

UAVs or consumer drones with computer vision algorithms are popularly used for object detection [11]. They are also used for remote sensing applications and traffic congestion recognition [12]. UAVs can also be used for cooperative communication [13], [14]. They can operate as amplify-and-forward (AF) relays [15]. By optimizing their power and trajectory, connectivity can be enhanced by minimizing the network outage probability. In [16], UAV relays assist a satellite in communicating with ground user equipment, and the outage probability is evaluated for opportunistic UAV relay selection. In [17], the use of UAVs as cooperative relays has been proposed, and the impact of increasing the number of intermediate relays and channel information rates on system reliability and outage probability has been investigated. This work uses DF scheme for relaying the data. However, the literature has not explored hybrid relaying schemes involving UAVs.

Recently, UAVs have also become popular for revolutionizing healthcare [18]. UAVs can deliver medications and test kits to remotely located patients (consumers) with chronic diseases [19]. An in-depth review of healthcare delivery using UAVs has been presented in [20]. Another important feature of UAVs is that they can act as communication relays for collecting patient data, processing it, and forwarding it for further diagnosis [21]. Using UAVs to relay patient data was found to save 20% of the time in urban areas and 60% time in remote areas [22]. The safe and secure communication of sensitive patient data using UAVs has been discussed in [23]. Drones have been used to detect COVID-19 patients based on thermal images [24]. However, the drones were used only for collecting data, which was analyzed using ML and DL on the cloud. UAVs and machine learning have been used for crop disease identification but not for healthcare [25]. UAV-assisted healthcare has been considered in [26] for COVID-19 detection. UAVs are relays for routing patient health data like body temperature, respiration rate, etc. The UAVs process and filter this data and forward it only if the symptoms are positive. However, ML or DL has not been applied to filter the data. Incorporating DL algorithms in UAVs, along with hybrid relaying and then accessing the outage probability, can be an important area of research.

B. Major Contributions

The major knowledge gap in the existing works is that they do not consider using consumer UAVs with multiple antennas for cooperative communication of patient data. Moreover, they do not incorporate the DL algorithms at the UAVs and access the outage probability of different relaying schemes. The proposed work considers UAVs with multi-antennas and DL-based relaying schemes for a smart healthcare system. Compared to a single-antenna UAV, a multi-antenna UAV can improve spectral efficiency, offering better service quality [27].

The following are the major contributions of the proposed work to the field of existing knowledge:

- In the context of cooperative communication utilizing a multi-antenna multi-UAV system, UAVs are employed to relay patient data by selecting the optimal UAV and the best antenna on the chosen UAV.
- A hybrid relaying scheme incorporating DL is proposed to achieve better performance.
- Patient data received at the UAV is decoded using DF. Each UAV is equipped with a pre-trained DNN to provide immediate feedback to the remote patient based on the decoded data. This approach facilitates rapid diagnosis and timely remedial actions by reducing the delay associated with transmitting the data to a medical expert at a hospital.
- Data is forwarded only if the DNN cannot diagnose accurately, in which case the data is amplified and forwarded to the doctor using the AF relaying scheme. Furthermore, the best transmit antenna at the UAV sends patient data to the doctor with the lowest possible outage probability.
- Using UAVs as edge devices is another important contribution of this work.

Thus, the work proposes a hybrid relaying scheme that combines AF and DF to achieve better performance and integrates a DL algorithm for efficient decision-making. The advantage of using a hybrid relaying scheme at the UAVs is to improve the QoS in the communication between the patient and the doctor, and the advantage of using the DL algorithm in UAVs is the faster response based on the information received from the patient.

The remaining work is prepared as follows: Section II describes the proposed cooperative communication network model for smart healthcare systems. Section III describes the proposed DL model and design approach. Section IV performs the Outage Probability (OP) of the considered network model, while Section V discusses the proposed network model results. Finally, Section VI concludes the paper.

II. DL-Based Smart Healthcare Using UAVs Assisted Cooperative Communication

UAVs are promising as the next-generation relaying devices for data generated from wearable electronic sensors. Multi-UAV cooperation can realize more effective sensing and transmission services and efficiently deliver the sensory data to the control center, for further analysis based on the received signal from wearable electronic devices.

As illustrated in Fig. 1, we consider the several patients', UAVs, and receivers for the study. The wearable devices at the patients' end send the health data to the health center using UAV-assisted cooperative communication. The communication system thus involves the patients, the non-collinearly distributed UAVs, and the healthcare center.

The wireless network model for smart health care consists of N senders (patients), M receivers (doctors), and K non-collinearly distributed UAVs between sender and receiver, as shown in Fig 2. The patient has several wearable electronic devices that take the bodily parameters and send them to the UAV.

Due to obstacles or long distances, we are not considering a direct link between the sender and the receiver, so these K UAVs act as relays between the sender and the receiver. Each terminal (sender, receiver, and UAVs) operates in a half-duplex mode, and all the UAVs use the proposed hybrid relaying protocol.

The transmission power of the link S to $UAV_k$is designated by ξ_S and ξ_{UAV_k} respectively. Channel matrix $N_S \times N_{UAVs_k}$ from S to $UAVs_k$ is designated by $H_{1,k}$, the component is $h_{1,k}(n_S, n_{UAVs_k})$ for $n_S = 1, 2, 3, \ldots, N_S$ and $n_{UAVs_k} = 1, 2, 3, \ldots, N_{UAVs_k}$; and the channel matrix $N_{UAVs_k}\ X\ N_R$ is designated by $H_{2,k}$,the component is $h_{2,k}(n_{UAVs_k}, n_R)$ for $n_{UAVs_k} = 1, 2, 3, \ldots, N_{UAVs_k}$ and$n_{UAVs_k} = 1, 2, 3, \ldots, N_R$.

We suppose that coefficients of all are fixed channels during all time slots; each component of $(H_{1,k})$ is independent and modeled as $h_{1,k}(n_S, n_{UAVs_k}) \sim CN(0, \Omega_{1,k})$ here $h \sim CN(0, \Omega)$ shows that h is a circularly symmetric complex-valued Gaussian random variable with zero mean and variance Ω; each component of $(H_{2,k})$ is independent and modeled as $h_{2,k}(n_{R_k}, n_S) \sim CN(0, \Omega_{2,k})$; and $(H_{1,k})$ and $(H_{2,k})$ are mutually independent, we use Binary Phase-shift keying (BPSK) modulation at the sender to transmit the data. Also, we assume that R knows all the channel matrices. $\{H_{1,k}, H_{2,k}{:}k = 1, 2, 3, \ldots, K\}$, and $UAVs_k$ knows the channel matrix $H_{1,k}$. As we considered BPSK modulation the noise in all channels is modeled as a mutually independent additive white Gaussian noise (AWGN) with zero mean and unit variance. TAS is the best method in traditional MIMO systems to develop multiple antennas with low feedback signaling overhead; it also exhibits tremendous performance with the highest diversity order [28], [29].

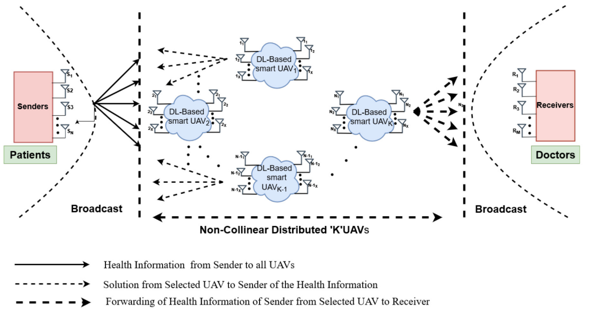

Fig. 2. UAV Assisted Cooperative Communication System for Smart Healthcare.

A. Combining Scheme

In this paper, switch and stay combining (SSC) is employed at the UAV receiver, as shown in Fig. 2. SSC is chosen over maximal ratio combining (MRC) and selection combining (SC) due to its enhanced performance and reduced complexity in diversity systems [30]. MRC's sensitivity to channel estimation errors, especially in low SNR conditions, limits its effectiveness. On the other hand, SC's performance is lower than that of MRC because it cannot fully exploit spatial diversity.SSC outperforms both by selecting a diversity path until its quality falls below a threshold, at which point it switches to another path. This approach minimizes power consumption, a critical consideration given the limited battery life of UAVs.

B. First Hop

In the first hop, the n_S^{th} patient broadcasts the health information to all UAVs. The UAV with the highest received SNR is selected as a relay based on the polling mechanism [31]. The instantaneous SNR of the SSC at UAV_k is given by $\gamma_{1,k}(n_S) = \xi_S |h_{1,k}(n_S, n_{UAV_k})|^2$.

Thus, this scheme selects the relaying UAV (UAV_k) by maximizing channel gain from the sender (S)to the UAV (UAV_k), which can be expressed as:

$$\psi_{S\rightarrow UAV_k} = \arg_{k=1,2,3,\ldots,k}^{\max} \gamma_{S\rightarrow UAV_K}^{x_2^{(1)}}, k \in \{1, 2, 3, \ldots, k\} \quad (1)$$

C. Second Hop

In the second hop, n_{UAV_k}-th antenna at $UAVs_K$ re-transmits the health information received in the first hop to the doctor using a hybrid relaying algorithm described in subsequent sections. At the receiver end, SSC is again used as the diversity scheme. The instantaneous SNR is $\gamma_{2,k}(n_{UAV_k})$ of SSC at the receiver (R) is $\gamma_{2,k}(n_{UAV_k}) = \xi_{UAV_k} |h_{2,k}(n_{UAV_k}, n_R)|^2$.

The received SNR at the doctor depends on the antenna used at UAV_k to re-transmit the message. Therefore, the best antenna at the relaying UAV (UAV_k) must be selected for optimal performance. This transmit antenna selection (TAS), along with SSC at the relaying UAV, is named as the joint UAV and TAS selection (JUTS) scheme.

We first define the maximum SNR value among all the output instantaneous SNRs ($\{\gamma_{1,k}(n_S)\}_{n_S=1}^{N_S}$) of SSC at UAV_k as $\Gamma_{1,k} \overset{def}{=} \max_{n_S=1,2,3,\ldots,N_S} \gamma_{1,k}(n_S)$. We also define the maximum SNR ($\{\gamma_{2,k}(n_{UAV_k})\}_{n_{UAV_k}=1}^{N_{UAV_k}}$) value among all the output instantaneous SNRs of SSC at R as $\Gamma_{1,k} \overset{def}{=} \max_{n_{UAV_k}=1,2,\ldots,.,N_{UAV_k}} \gamma_{2,k}(n_{UAV_k})$. Note that $\Gamma_{1,k}$ and $\Gamma_{2,k}$ can be obtained in first hop ($S \rightarrow UAV_k$) and second hop ($UAV_k \rightarrow R$) respectively. Therefore, considering network adapting TAS at the sender and SSC at the receiver, $\Gamma_{1,k}$ and $\Gamma_{2,k}$ are equivalent SNRs of the first hop and second hop, respectively. Here we sort the equivalent SNRs $\{\Gamma_{2,k}\}_{k=1}^{K}$ in a cascade order as: $\Gamma_{2,\langle 1\rangle} \geq \Gamma_{2,\langle 2\rangle} \geq, \ldots, \geq \Gamma_{2,\langle K\rangle}$, where $\Gamma_{2,\langle k\rangle}$ indicates the k^{th} highest equivalent SNR in second hop. Further, $UAV_{\langle k\rangle}$ denotes a UAV terminal linked with $\Gamma_{2,\langle k\rangle}$. Note that $\Gamma_{1,\langle k\rangle}$ is not the k^{th} largest SNR of the first hop; but it is the highest SNR of the first hop, which is linked with the k^{th} highest equivalent SNR, $\Gamma_{2,\langle k\rangle}$, of the second hop. Here, we determine the best transmit antenna at S, one best UAV, and the best transmit antenna at the selected UAV.

III. Outage Probability of the Proposed Scheme

This section discusses the OP of the proposed scheme that uses the JUTS selection scheme and the hybrid relaying algorithm incorporating DL.

A. Outage Probability of JUTS

The OP of the JUTS is given in the lemma below.

Lemma 1: The JUTS OP is given by

$$P_{out}^{UAV-TAS}(UAV) = \psi_1(UAV; K) + \psi_2(UAV; K) \tag{2}$$

Where

$$\psi_1(UAV; K) = \sum_{l=1}^{K}\left(\prod_{k=1}^{l-1} P_r\left[\Gamma_{1,\langle k\rangle} < \bar{UAV}\right]\right) \times P_r[\Gamma_{1,\langle l\rangle} \geq \bar{UAV}]P_r[\Gamma_{2,\langle l\rangle} < \bar{UAV}], \tag{3}$$

$$\psi_2(UAV; K) = \prod_{k=1}^{K} P_r\left[\Gamma_{1,\langle k\rangle} < \bar{UAV}\right] \tag{4}$$

Proof: The outcomes of the JUTS scheme can be divided into two cases;

case1: One UAV is selected out of N UAVs

case2: No UAV is selected

In case 1, if the link from UAVk to R is in the outage, i.e., $\Gamma_{2,\langle \bar{k}\rangle} < \bar{UAV}$, then the network is in outage. With the help of equation (2), we can obtain equation (10). In case 2, since all the links from S to $\{UAV_k\}_{k=1}^{K}$ are in the outage, it is noticeable that the network is in the outage. Therefore, equation (11) can be obtained. Then adding equations (10) and (11) yields question (9). Here we can notice that $\psi_1(UAV; K)$ is the total OP when one UAV is selected, and $\psi_2(UAV; K)$ is the total OP, then no UAV is selected. Also, from (10) $\psi_1(UAV; K)$ and from (11) $\psi_2(UAV; K)$ can be recursively calculated as follows:

$\psi_1(UAV; K) = \psi_2(UAV; K-1)P_r[\Gamma_{1,\langle k\rangle} \geq \bar{UAV}]P_r[\Gamma_{2,\langle k\rangle} < \bar{UAV}] + \psi_1(UAV; K-1)\psi_2(UAV; K) = \psi_2(UAV; K-1)P_r[\Gamma_{1,\langle k\rangle} < \bar{UAV}]$

Where $\psi_1(UAV; 0) = 0$ and $\psi_2(UAV; 0) = 1$ are the initial conditions of $\psi_1(UAV; 0)$ and $\psi_2(UAV; 0)$ respectively.

The closed-form expression OP of JUTS is possible to obtain when we directly solve (9) with order statistics [32]. However, using this approach is extremely lengthy and complicated. Therefore, we use another useful and more understanding approach in this study. We obtain the OP of JUTS from Lemma 2 that the OP $P_{out}^{UAV-TAS}(UAV)$ does not need to be directly solved. Rather, we can use expression (7) of [33] for JUTS, which achieves full diversity order.

The proposed scheme also involves a hybrid relaying algorithm that affects the outage probability, which is described in the next subsection.

B. Hybrid Relaying Algorithm

The proposed hybrid relaying algorithm utilizes UAVs equipped with pre-trained DNNs to enhance the performance and reliability of transmitting patient data. Initially, patient data received at the UAV is decoded using the DF scheme. The decoded data is then processed by the DNN, which provides immediate feedback if it can accurately diagnose the patient's condition, thereby facilitating rapid diagnosis and timely remedial actions without the delay of transmitting data to a distant medical expert. If the DNN cannot diagnose accurately, the UAV switches to the AF relaying scheme, amplifying the received signal and forwarding it directly to a doctor. The UAV employs the best available transmit antenna to ensure the highest reliability and lowest outage probability during transmission. Combining DF, AF, and deep learning, this hybrid approach optimizes resource utilization, reduces delays, and enhances the rapid response capability in remote medical diagnostics using UAVs. The hybrid relaying algorithm using the DNN is illustrated in Algorithm 1.

IV. DNN Used in the Proposed Scheme

The proposed hybrid relaying scheme assists the sender by transferring the data to the receiver based on the decision taken by the pre-trained DNN at the UAV. The DNN in the UAV is a pre-trained model trained offline and loaded into the UAV. Onboard training of the DNN may have advantages, but loading it with a pre-trained DNN is an efficient option because the UAV operates on a limited battery. Once trained, the disease is predicted based on the patient's symptoms. The patient data is amplified and forwarded to the healthcare expert/receiver only when the decision accuracy is below the threshold, which means that the UAV cannot accurately predict the disease, so the expert's opinion is needed to confirm the disease. A more detailed description of the DNN's architecture is given in this section.

Algorithm 1 Hybrid Relaying Using the DL Algorithm

1. Input: Data set, decision threshold (d_t)

2. Output: Predict decision type

3. Network: The hidden layer, the number of neurons per layer, the activation, the kernel, the learning rate, the epochs, and the output functions of the hidden and output layers should all be set.

4. **for** S=1:N; UAV=1:K:

Find $\gamma_{1,k}(n_s) = \xi_s \left|h_{1,k}(n_s, n_{UAV_k})\right|^2$

5. Select UAV with the highest $\gamma_{1,k}(n_s)$ as UAV_k

6. Decode the data at UAV_k using the DF scheme and use the pre-trained DL model to diagnose using the decoded data.

7. **If** decision accuracy $\geq d_t$, make decision and response with a solution to S ($UAV_k \rightarrow S$)

8. **If** decision accuracy $< d_t$, amplify the decoded data and send it to the doctor using AF relaying scheme ($UAV_k \rightarrow S$) and the TAS

$$\gamma_{2,k}(n_{UAV_k}) = \xi_{UAV_K} \left|h_{2,k}(n_{UAV_k}, n_R)\right|^2$$

9. Return the predicted information with a solution to UAV_K ($R \rightarrow UAV_K$)

10. Then UAV_K receives the solution from R and forwards it to S ($R \rightarrow UAV_K \rightarrow S$)

$$\gamma_{1,k}(n_s) = \xi_s \left|h_{k,1}(n_{UAV_k}, n_S)\right|^2$$

end

A. Data Set Preparation

Medical data preprocessing is crucial for the success of machine learning models [34], [35]. It involves steps such as data cleaning, transforming, and normalization, etc. For the training and prediction of the proposed DNN, an existing patient data set consisting of 1041 samples has been taken. Based on the seven different symptoms: temperature, pulse rate, lower abdomen pain, upper abdomen pain, vomiting feeling, yellowish urine, and indigestion, it will be decided whether the patient has heart disease, viral fever, jaundice, food poisoning, or is in a healthy condition [36].Each patient's data is converted into a bit stream, with eight bits representing each symptom, resulting in 56 bits per patient. This bit stream is then BPSK modulated and transmitted across a range of Signal-to-Noise-Ratio (SNR) values, from -10 to 20 dB. Samples with missing data were removed, and the remaining data was normalized. The dataset was also resampled to ensure equal representation of each class, resulting in a total of 25,452 samples. Of these, 20615 samples (81%) were used for training the DNN, 2290 (9%) were used for validation, and the remaining 2,547 samples (10%) were used for testing. In machine learning, training data is used to teach the model by allowing it to learn patterns and relationships. Validation data helps tune the model's parameters and prevents overfitting by evaluating its performance during training. Finally, testing data is used to provide an unbiased assessment of the model's ability to generalize to new, unseen data. The properties of the communication system are shown in Table I.

TABLE I
MONTE CARLO SIMULATION PARAMETERS

Symbol	Description	Value
N	Number of bits	10^6
α	Path loss factor	3
n	Noise	AWGN
h	Channel	Rayleigh fading
Mod	Modulation	BPSK
SNR	Transmitting SNR	-10dB to 20dB
SNRth	SNR threshold at selected UAV	$\text{SNR}_{S \rightarrow UAV_k}$ (variable)
S	Position of Source(patient)	0m
R	Position of Reciever (Doctor)	10000m
UAVs	Position of UAVs	In between patient and doctor

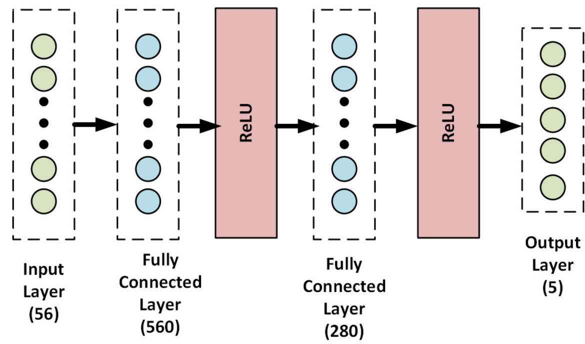

Fig. 3. Proposed architecture of DNN.

B. DNN Architecture

The proposed framework for implementation of DNN design in all UAVs includes the training phase and decision-making phase, and the proposed DL design learns the decision-making based on the received information from the sender respectively. In the training phase, the DNN is trained with the parameters of different health symptoms received from the sender. In the decision-making phase, consumer electronic device UAVs make the decision based on the trained parameters in the UAV. The proposed DNN architecture is shown in Fig. 3.

The proposed DNN consists of two fully connected hidden layers. The first hidden layer has 560 neurons, and the second has 280 neurons. The training and the validation accuracy are 99.6% and 96.6%, respectively. Increasing the number of hidden layers beyond 2 reduced the model's accuracy. Even though the training accuracy increased to 99.8%, the validation accuracy was slightly reduced to 95.8%, indicating overfitting. Similarly, when the number of neurons increases beyond 840 (560 in the first layer and 280 in the second layer, the validation accuracy slightly reduces, indicating model overfitting. Thus, a DNN with two hidden layers containing 560 neurons in the first and 280 neurons in the second was chosen. The details about the DNN are shown in Table II. It should be noted that the proposed DNN architecture cannot give the best results for all data types and needs to be modified based on the patient data. However, using an appropriate DNN, the proposed approach based on cooperative communication and relaying by the consumer electronics devices UAVs assisted cooperative communication can be generalized to any other patient data. The DNN was trained for 8 epochs. The number of epochs was taken based

TABLE II
DL SIMULATION PARAMETERS

Parameters	Values
Number of samples taken for the training dataset	22905
Number of the test dataset	2547
Activation function	ReLU
Loss function	Categorical Crossentropy
Optimizer	Adam
Number of epochs	8
Number of hidden layers	2
No. of neurons in hidden layer 1	560
No. of neurons in hidden layer 2	280
Batch-size	50

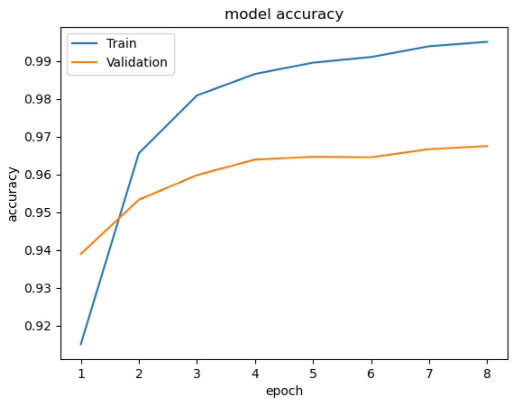

Fig. 4. Model accuracy vs epochs with training and validation data.

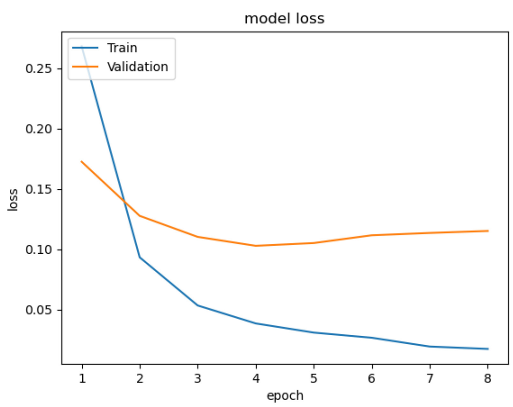

Fig. 5. Model loss vs epochs with training and validation data.

on the model accuracy and loss curves to avoid overfitting the model. The training and validation accuracy curves are shown in Fig. 4 and the corresponding loss curves are shown in Fig. 5.

The confusion matrix with the training and validation data is shown in Fig. 6 and that with the test data is shown in Fig. 7. The average precision of the model with the training and validation data is 0.9972, and its average recall is 0.9972. With the test data, the accuracy is 96.62%, the average precision is 0.9658, and the average recall is 0.9659. Thus, these high values of precision and recall indicate the model is good at classifying the patient data. From the accuracy and loss curves, the confusion matrix plots, precision and recall scores, it can be concluded that the model can accurately predict the disease based on the patient data.

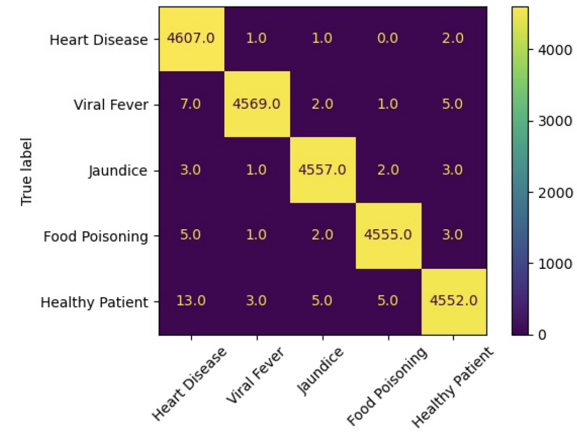

Fig. 6. Confusion matrix for DL model with training and validation data.

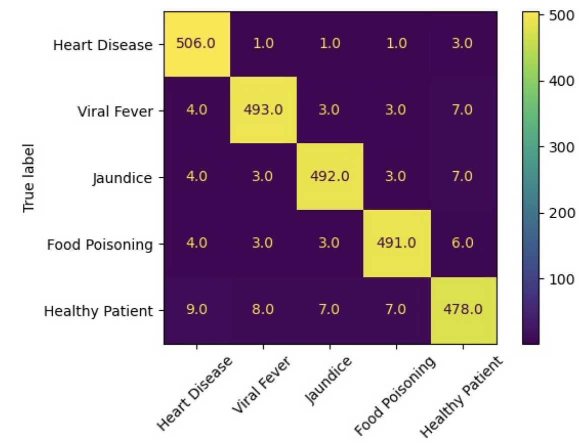

Fig. 7. Confusion matrix for DL model with the testing data.

V. RESULTS AND DISCUSSION

A. Outage Probability Estimation

Monte Carlo simulation is used to simulate the network. Multiple multi-antenna UAVs have been considered between sender and receiver; the location of S, R and UAV_K ($K = 1, 2, 3, \ldots, k$) are (0,0), (0,10000) and non-collinearly distributed K UAVs in between S and R respectively. The path loss factor is taken as 3. Fig. 8 illustrates the throughput of different UAV relaying schemes with the proposed UAV relaying scheme.

The end-to-end system performance is the lowest, and the OP is larger when the selected AF-based UAV is installed with a single antenna $K_{UAV_k} = 1$,$N_S = 1$, and $M_R = 1$. On the other hand, when the number of antennas on the chosen UAV increases, this scheme's OP rapidly decreases, and the minimal OP of this scheme is given at $K_{UAV_k} = 4$, $N_S = 1$, and $M_R = 1$.

Comparatively to the AF-based UAV scheme, the DF-based UAV scheme gives a smaller OP for all the scenarios such $k = 1, 2, 3, 4$, and 5 as shown in Figs. 9, 10, and 11 which show the effect of the OP on the two types of UAV schemes. In contrast, the suggested hybrid-based UAV scheme experiences significantly fewer outages than AF-based and DF-based UAV systems. When the hybrid-based UAV is deployed with

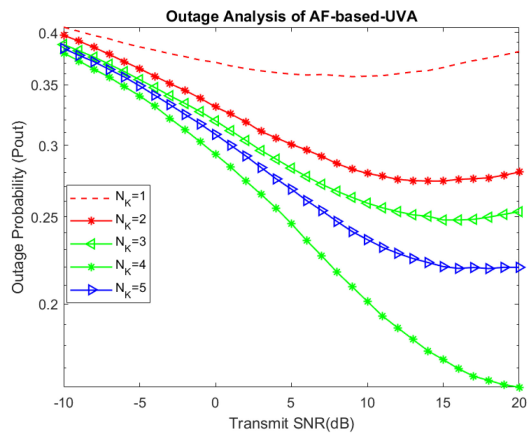

Fig. 8. The effect of outage probability for $k = 1, 2, 3, 4$, and 5, on AF-based.

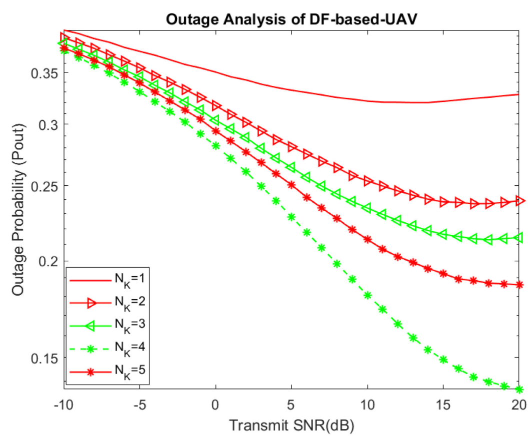

Fig. 9. The effect of outage probability for $k = 1, 2, 3, 4$, and 5 on DF-based.

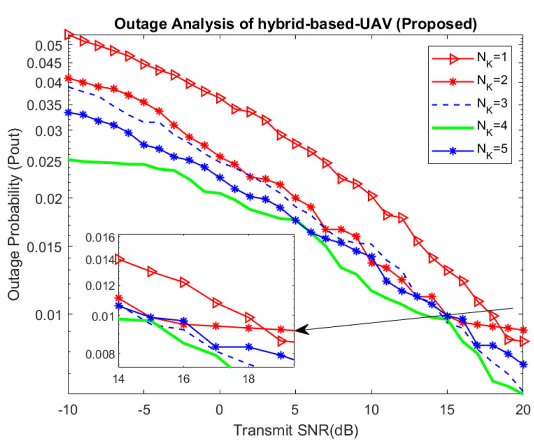

Fig. 10. The effect of outage probability for $k = 1, 2, 3, 4$, and 5 on Hybrid-based consumer electronic device UAV.

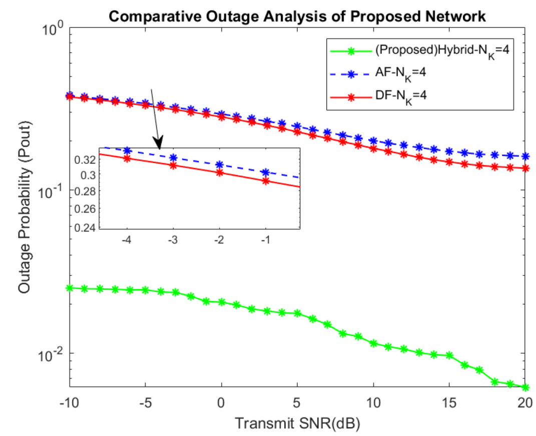

Fig. 11. Comparative outage probability analysis of proposed hybrid-based UAV with AF-based and DF-based for $k = 4$.

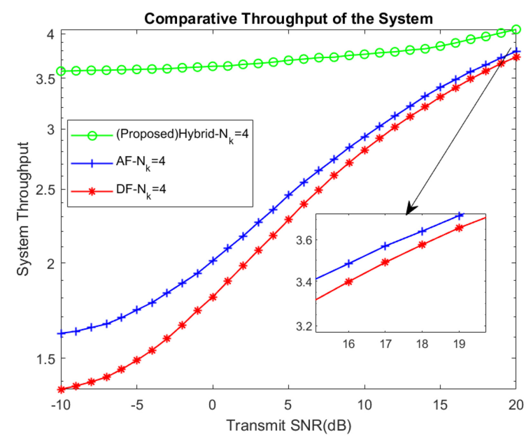

Fig. 12. Comparative system throughput analysis of proposed hybrid relaying scheme with AF-based and DF-based UAVs for $k = 4$.

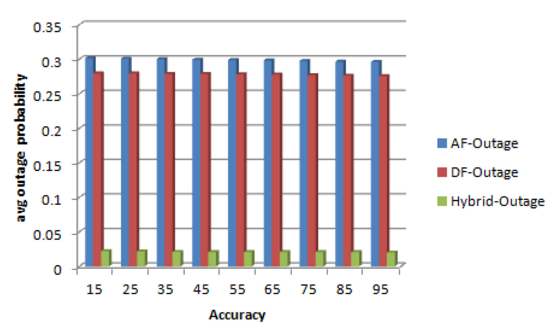

Fig. 13. Accuracy vs. comparison of Outage Probability for Different Relaying Protocols.

$k = 1, 2, 3, 4$ in particular, it performs better than the other two systems.

The comparative OP analysis of the hybrid-based, AF-based, and DF-based UAV schemes is shown in Figure 12. In addition, we evaluated situations with $k = 5, 6, 7, 8, 9$, and 10 and found that the number of antennas mounted at UAV $k = 5$ offers outstanding performance for all the systems. But the OP is greater than $k = 4$ when we equipped $k = 6, 7, 8, 9$, and 10. We only considered $k = 4$ because having $k > 4$ antennas increases the amount of interference directed at the UAV. We evaluated all three methods for relative performance analysis, and it is clear that the suggested hybrid-DL-based UAV scheme outperforms the other two. Accuracy vs. comparison of outage probability for different relaying protocols is shown in Fig. 13.

VI. Conclusion

The deep learning-based multiple consumer electronics devices UAVs assisted cooperative smart healthcare system for NTN, which consists of N senders, M receivers, and K non-collinearly distributed UAVs with multiple antennas, between sender and receiver, was examined in this study. UAVs act as relaying stations between the patient and the healthcare facility. UAVs use the DL model to diagnose the patient based on the data sent by him. If unable to decide with the desired accuracy, they forward the data to the expert at the healthcare facility. Various UAV and transmit antenna selection strategies were presented to enhance the performance of the remote user. Optimum UAV-TAS has been selected and a hybrid relaying scheme has been investigated in terms of the outage probability. The proposed hybrid-DL-based scheme outperforms AF-based and DF-based schemes to decrease the outage probability and provide a reliable healthcare service to the remote user. The authors intend to consider the energy efficiency of the proposed schemes and incorporate the limited battery power of the UAV to provide reliable service to remote patients. Another, important limitation and future research is to model communication between the UAVs and to develop a more generic DL model for predicting diseases based on patient data.

REFERENCES

[1] Y. Huo, X. Fan, L. Ma, X. Cheng, Z. Tian, and D. Chen, "Secure communications in tiered 5G wireless networks with cooperative jamming," *IEEE Trans. Wireless Commun.*, vol. 18, no. 6, pp. 3265–3280, Jun. 2019.

[2] J. Gruner, A. Bloße, M. Rockstroh, T. Neumuth, and P. Rostalski, "Requirement analysis for an aerial relay in emergency response missions," *Current Directions Biomed. Eng.*, vol. 6, no. 3, pp. 16–19, 2020.

[3] S. P. Mohanty, "Unmanned arial vehicles as consumer electronics [notes from the editor]," *IEEE Consum. Electron. Mag.*, vol. 8, no. 3, pp. 3–4, May 2019, doi: 10.1109/MCE.2019.2892216.

[4] N. Kumar, D. Puthal, T. Theocharides, and S. P. Mohanty, "Unmanned aerial vehicles in consumer applications: New applications in current and future smart environments," *IEEE Consum. Electron. Mag.*, vol. 8, no. 3, pp. 66–67, May 2019.

[5] L. Daoud and A. El-Mahdy, "Performance of hybrid relaying in cooperative wireless systems in the presence of impulsive noise," in *Proc. Signal Process. Algorithms, Archit., Arrang., Appl. (SPA)*, 2014, pp. 154–159.

[6] K. Tezuka, K. Maruta, and C.-J. Ahn, "Variable channel threshold-based hybrid AF-DF cooperative communications with packet splitting," in *Proc. IEEE Int. Conf. Consum. Electron.-Asia (ICCE-Asia)*, 2018, pp. 206–212.

[7] S. Malik et al., "Hybrid multihop relay protocol using dynamic power allocation for LTE-advanced system," *Wireless Pers. Commun.*, vol. 77, pp. 2945–2958, Aug. 2014.

[8] X. Song, M. Zhang, W. Liu, and F. Liu, "Threshold-based hybrid relay selection scheme," in *Proc. 12th World Congr. Intell. Control Autom. (WCICA)*, 2016, pp. 222–227.

[9] F. A. Onat, A. Adinoyi, Y. Fan, H. Yanikomeroglu, J. S. Thompson, and I. D. Marsland, "Threshold selection for SNR-based selective digital relaying in cooperative wireless networks," *IEEE Trans. Wireless Commun.*, vol. 7, no. 11, pp. 4226–4237, Nov. 2008.

[10] L. Kang and Y. Qi, "A relay selection strategy based on network fairness under hybrid duplex mode," in *Proc. Cross Strait Quad-Reg. Radio Sci. Wireless Technol. Conf. (CSQRWC)*, 2019, pp. 1–3.

[11] Y. Zhang, W. Liang, X. Yuan, S. Zhang, G. Yang, and Z. Zeng, "Deep learning-based abnormal behavior detection for elderly healthcare using consumer network cameras," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 2414–2422, Feb. 2024.

[12] T. K. Behera, S. Bakshi, M. A. Khan, and H. M. Albarakati, "A lightweight multiscale-multiobject deep segmentation architecture for UAV-based consumer applications," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 3740–3753, Feb. 2024.

[13] S. Javaid et al., "Communication and control in collaborative UAVs: Recent advances and future trends," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 6, pp. 5719–5739, Jun. 2023.

[14] Y. Li, Y. Bi, J. Wang, Z. Li, H. Zhang, and P. Zhang, "Unmanned aerial vehicle assisted communication: Applications, challenges, and future outlook," *Clust. Comput.*, vol. 27, pp. 13187–13202, Dec. 2024.

[15] H. S. Khallaf, S. Hashima, M. Rihan, E. M. Mohamed, and H. M. Kasem, "Quantifying impact of pointing errors on secrecy performance of UAV-based relay-assisted FSO links," *IEEE Internet Things J.*, vol. 11, no. 2, pp. 2979–2989, Jan. 2024.

[16] P. K. Sharma, D. Deepthi, and D. I. Kim, "Outage probability of 3-D mobile UAV relaying for hybrid satellite-terrestrial networks," *IEEE Wireless Commun. Lett.*, vol. 24, no. 2, pp. 418–422, Feb. 2020.

[17] Y. Zeng, R. Zhang, and T. J. Lim, "Throughput maximization for UAV-enabled mobile relaying systems," *IEEE Trans. Commun.*, vol. 64, no. 12, pp. 4983–4996, Dec. 2016.

[18] A. Merei, H. Mcheick, and A. Ghaddar, "Survey on path planning for UAVs in healthcare missions," *J. Med. Syst.*, vol. 47, no. 1, p. 79, 2023.

[19] S. J. Kim, G. J. Lim, J. Cho, and M. J. Côté, "Drone-aided healthcare services for patients with chronic diseases in rural areas," *J. Intell. Robot. Syst.*, vol. 88, pp. 163–180, Oct. 2017.

[20] H. A. Campbell et al., "Emerging research topics in drone healthcare delivery," *Drones*, vol. 8, no. 6, p. 258, 2024.

[21] S. Ullah et al., "UAV-enabled healthcare architecture: Issues and challenges," *Future Gener. Comput. Syst.*, vol. 97, pp. 425–432, Aug. 2019.

[22] A. Sharma, P. K. Singh, P. Nikashina, V. Gavrilenko, A. Tselykh, and A. Bozhenyuk, "Role of UAVs in delivering the patient samples from one hospital to different hospital," in *IoT, Big Data and AI for Improving Quality of Everyday Life: Present and Future Challenges*. Cham, Switzerland: Springer, 2023, pp. 139–152.

[23] S. Pirbhulal, W. Wu, G. Li, and A. K. Sangaiah, "Medical information security for wearable body sensor networks in smart healthcare," *IEEE Consum. Electron. Mag.*, vol. 8, no. 5, pp. 37–41, Sep. 2019.

[24] A. Kumar, K. Sharma, H. Singh, S. G. Naugriya, S. S. Gill, and R. Buyya, "A drone-based networked system and methods for combating coronavirus disease (COVID-19) pandemic," *Future Gener. Comput. Syst.*, vol. 115, pp. 1–19, Feb. 2021.

[25] A. Abbas et al., "Drones in plant disease assessment, efficient monitoring, and detection: A way forward to smart agriculture," *Agronomy*, vol. 13, no. 6, p. 1524, 2023.

[26] A. Çalhan and M. Cicioğlu, "Drone-assisted smart data gathering for pandemic situations," *Comput. Elect. Eng.*, vol. 98, Mar. 2022, Art. no. 107769.

[27] J. Zhang, Y. Zeng, and R. Zhang, "Multi-antenna UAV data harvesting: Joint trajectory and communication optimization," *J. Commun. Inf. Netw.*, vol. 5, no. 1, pp. 86–99, Mar. 2020.

[28] P. Yang, Y. Xiao, Y. L. Guan, S. Li, and L. Hanzo, "Transmit antenna selection for multiple-input multiple-output spatial modulation systems," *IEEE Trans. Commun.*, vol. 64, no. 5, pp. 2035–2048, May 2016.

[29] L. Cao, X. Zhang, Y. Wang, and D. Yang, "Transmit antenna selection strategy in amplify-and-forward MIMO relaying," in *Proc. IEEE Wireless Commun. Netw. Conf.*, 2009, pp. 1–4.

[30] M. K. Jataprolu, D. S. Michalopoulos, and R. Schober, "Colocated and distributed switch-and-stay combining: Optimality under switching rate constraints," *IEEE Trans. Veh. Technol.*, vol. 63, no. 1, pp. 451–457, Jan. 2014.

[31] C. Yingnan, X. Li, and L. Kai, "Research on node selection algorithm of near space relay link," in *Proc. 5th Int. Conf. Commun. Eng. Technol. (ICCET)*, 2022, pp. 27–31.

[32] U. Kamps, "A concept of generalized order statistics," *J. Statist. Plann. Inference*, vol. 48, no. 1, pp. 1–23, 1995.

[33] M. Ju, H.-K. Song, and I.-M. Kim, "Joint relay-and-antenna selection in multi-antenna relay networks," *IEEE Trans. Commun.*, vol. 58, no. 12, pp. 3417–3422, Dec. 2010.

[34] I. Chahid, A. K. Elmiad, and M. Badaoui, "Data preprocessing for machine learning applications in healthcare: A review," in *Proc. 14th Int. Conf. Intell. Syst., Theor. Appl. (SITA)*, 2023, pp. 1–6.

[35] A. Idri, H. Benhar, J. Fernández-Alemán, and I. Kadi, "A systematic map of medical data preprocessing in knowledge discovery," *Comput. Methods Programs Biomed.*, vol. 162, pp. 69–85, Aug. 2018.

[36] "Disease predictor." 2019. [Online]. Available: https://www.kaggle.com/datasets/edgarjames/disease-predictor